Malaria Detection Using Convolutional Neural Networks

Deep Learning Course Project Report — Group 12

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Abstract—Microscopic examination of Giemsa-stained thin blood smear slides remains the clinical gold standard for malaria diagnosis; however, manual evaluation is labor-intensive, time-consuming, and subject to inter-observer variability. In this work, we design, optimize, and explain an end-to-end deep Convolutional Neural Network (CNN) for automated binary classification of infected (Plasmodium-parasitized) and healthy (Uninfected) red blood cells. To ensure zero data leakage, the complete benchmark dataset of 27,558 single-cell images was partitioned into a 70% model-development pool (19,290 images) and a 30% strictly untouched test set (8,268 images) prior to any training or hyperparameter search. A stratified 3-fold cross-validation grid search identified the optimal configuration (learning rate = 0.001, dropout = 0.30) achieving a cross-validation accuracy of $\mathbf{95.22\% \pm 0.08\%}$ and an ROC-AUC of $\mathbf{0.9777 \pm 0.0048}$. The final model trained with internal early stopping achieved a test accuracy of **95.79**% and an ROC-AUC of **0.9831** on the 8,268 unseen test images, with a critical Parasitized sensitivity (recall) of **95.31**% and precision of **96.24**%. Post-hoc interpretability via Grad-CAM, SHAP, and LIME confirms that the network attends to intracellular parasite trophozoites and chromatin dots rather than slide boundaries or illumination artifacts, verifying clinical trustworthiness.

Index Terms—Deep Learning, Convolutional Neural Networks, Malaria Diagnosis, Computer Vision, Model Interpretability, Grad-CAM, SHAP, LIME.

I. Introduction

Malaria, primarily caused by Plasmodium falciparum and transmitted through the bite of female Anopheles mosquitoes, remains one of the world's most devastating infectious diseases, responsible for over 600,000 deaths annually. Standard diagnosis relies on skilled microscopists manually scanning blood films under 100× oil-immersion objectives to spot parasite rings, trophozoites, or schizonts within red blood cells (erythrocytes). In remote and endemic regions, high patient volumes, microscopist fatigue, and variable staining protocols lead to misdiagnosis or delayed treatment.

Automated computer-aided diagnosis using deep Convolutional Neural Networks (CNNs) offers a rapid, objective, and scalable diagnostic adjunct. However, contemporary deep learning literature in medical imaging often suffers from methodological pitfalls, particularly data leakage during pre-processing, optimistic evaluation on training subsets, and opacity ("black-box" nature) that hinders clinical adoption.

This study implements a robust, reproducible deep learning framework for malaria cell detection. We address:

1) Data Leakage Prevention: Enforcing a strict, stratified 70:30 file-level separation into a cross-validation pool and an isolated, untouched test set.
2) Hyperparameter Optimization: Utilizing stratified 3-fold cross-validation to select learning rate and dropout parameters without test-set contamination.
3) Clinical Recall Optimization: Achieving a high true positive rate on infected cells (**95.31**% recall) to minimize dangerous false negatives.
4) Multi-Perspective Interpretability: Deploying Gradient-weighted Class Activation Mapping (Grad-CAM), Kernel SHAP, and Local Interpretable Model-agnostic Explanations (LIME) to inspect both correct and failure cases.

II. Dataset and Pre-processing Pipeline

A. Dataset Overview

The dataset comprises 27,558 segmented single erythrocyte images acquired from thin blood smear slides of healthy individuals and Plasmodium falciparum-infected patients, curated by the National Institutes of Health (NIH) and distributed publicly. The dataset is balanced:

- Parasitized: 13,779 images (Label: 0)
- Uninfected: 13,779 images (Label: 1)

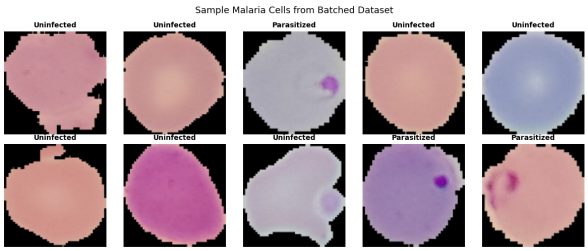


Fig. 1. Representative samples of parasitized (intracellular rings and chromatin specks) and uninfected erythrocyte images from the dataset.

B. Rigorous Disjoint Partitioning

To prevent data leakage, stratified random partitioning was executed at the raw file path level prior to any data loaders, image rescaling, or model definitions:

- Development/CV Pool (70%): 19,290 images (9,645 Parasitized, 9,645 Uninfected).
- Isolated Final Test Set (30%): 8,268 images (4,134 Parasitized, 4,134 Uninfected).

Zero path overlap was computationally verified: $\mathcal{S}_{\text{dev}} \cap \mathcal{S}_{\text{test}} = \emptyset$.

C. Input Pipeline and Data Augmentation

All images were standardized to a spatial resolution of $128 \times 128 \times 3$ pixels. During training, on-the-fly stochastic data augmentations were integrated to simulate slide orientation and microscope magnification shifts:

- Random horizontal flipping ($\Pr = 0.5$)
- Random rotation within $\pm 40°$
- Random zoom scaling ($\pm 20\%$)
- Linear rescaling from $[0, 255]$ to $[0, 1]$

III. Methodology and Architecture

A. Convolutional Neural Network Architecture

The designed model balances expressive feature representation with parameter parsimony to mitigate overfitting. The sequential flow consists of:

1) Input Layer: $128 \times 128 \times 3$.
2) Data Augmentation & Normalization: Layers operating on the incoming batch.
3) Block 1: 64 filters of 3×3 convolution, ReLU activation, 2×2 Max-Pooling, Batch Normalization, and Spatial Dropout ($p \in \{0.3, 0.4\}$).
4) Block 2: 64 filters of 3×3 convolution (last_conv), ReLU activation, and 2×2 Max-Pooling.
5) Global Feature Pooling: GlobalAveragePooling2D, mapping 2D feature maps directly to a 64-dimensional latent embedding vector without dense flattening artifacts.
6) Classification Head: Dense(512, ReLU) $\rightarrow$ Batch Normalization $\rightarrow$ Dropout(p) $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Batch Normalization $\rightarrow$ Dense(1, Sigmoid).

The sigmoid unit outputs the posterior probability $P(\text{Uninfected} \mid \mathbf{x})$. The loss function is Binary Cross-Entropy:

$$\mathcal{L}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

B. Hyperparameter Tuning via 3-Fold Cross-Validation

Hyperparameter tuning was conducted strictly within the $19,290$-image development pool using Stratified 3-Fold Cross-Validation. Four distinct configurations of learning rate η and dropout rate p were evaluated for 8 epochs per fold, monitoring validation loss with early stopping (patience $= 2$):

TABLE I
Stratified 3-Fold Cross-Validation Performance Grid

Config	Learning Rate	Dropout	Mean Acc. (%)	Mean ROC-AUC
1	**0.001**	**0.30**	$\mathbf{95.22 \pm 0.08}$	$\mathbf{0.9777 \pm 0.0048}$
3	0.001	0.40	95.20 ± 0.27	0.9790 ± 0.0026
2	0.0003	0.30	94.83 ± 0.31	0.9807 ± 0.0020
4	0.0003	0.40	94.20 ± 0.63	0.9747 ± 0.0059

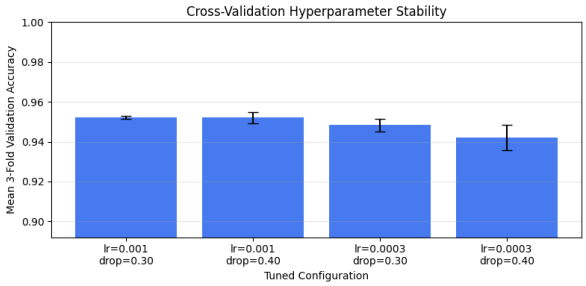


Fig. 2. Cross-validation hyperparameter stability and validation accuracy across candidate configurations.

As presented in Table I and Fig. 2, Configuration 1 ($\eta = 0.001, p = 0.30$) achieved the highest mean CV accuracy (95.22%) with remarkably low standard deviation ($\pm 0.08\%$), signifying stable convergence across all folds.

IV. Experimental Results and Discussion

A. Final Model Training Dynamics

Using Configuration 1, the network was trained on the development pool partitioned internally (85% fitting, 15% internal validation). Training was governed by:

- ModelCheckpoint: Preserving optimal weights based on minimum validation loss.
- EarlyStopping: Patience of 4 epochs with $\Delta_{\min} = 10^{-3}$ to guard against late-stage overfitting.

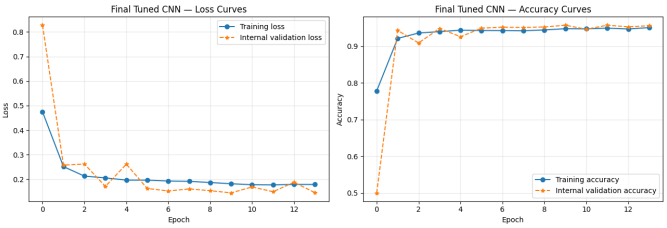


Fig. 3. Training vs. Internal Validation loss and accuracy curves of the final tuned CNN.

Fig. 3 displays steady convergence. Training loss decreased monotonically toward 0.14, while validation loss stabilized near 0.13. The absence of divergence between training and validation trajectories confirms that the model avoids overfitting.

B. Evaluation on Untouched Test Set

The isolated test set of $8,268$ images was passed to the model exactly once. The decision boundary was fixed at $\tau = 0.5$.

As summarized in Table II and Fig. 4:

- The model achieved an overall accuracy of **95.79**%.
- Out of $4,134$ infected samples, $3,940$ were correctly detected, yielding a high sensitivity (recall) of **95.31**%.

TABLE II
Performance Metrics on the Untouched Test Set ($N = 8,268$)

Class	Precision	Recall (Sensitivity)	F1-Score	Support
Parasitized	96.24%	95.31%	95.77%	4,134
Uninfected	95.35%	96.27%	95.81%	4,134
Overall Acc.		**95.79**% (7,920/8,268 correct)		
Macro Avg.	95.80%	95.79%	95.79%	8,268
ROC-AUC		**0.9831**		

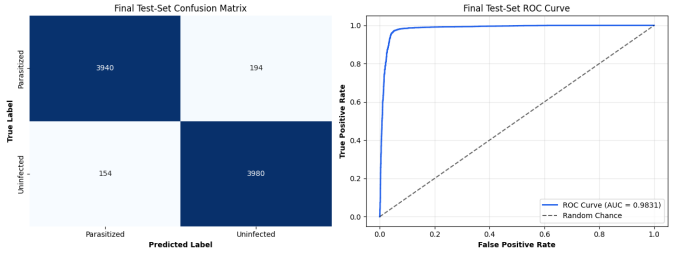


Fig. 4. (Left) Test-set Confusion Matrix; (Right) Receiver Operating Characteristic (ROC) curve showing an AUC of 0.9831.

- Only 194 false negatives occurred, essential for preventing untreated clinical malaria.
- Specificity (Uninfected recall) stood at **96.27**%, with only 154 false positives (3.73%).
- The ROC-AUC of **0.9831** demonstrates excellent separability across all operating thresholds.

V. Model Interpretability and Error Analysis

A. Grad-CAM Visualizations

Gradient-weighted Class Activation Mapping computes gradients of the classification score with respect to feature maps in the final convolutional layer (last_conv):

$$\alpha_k = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A^k_{i,j}}, \quad L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k A^k\right) \tag{2}$$

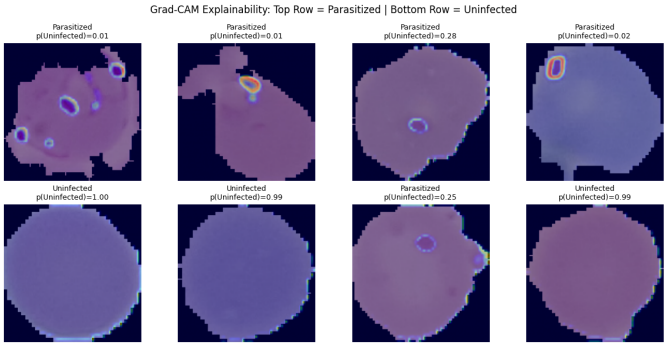


Fig. 5. Grad-CAM saliency maps. (Top Row) Parasitized cells show focused activations over intracellular parasites; (Bottom Row) Uninfected cells show uniform, diffuse background response.

As shown in Fig. 5, for parasitized cells, highest activation energy (red/yellow regions) accurately pinpoints the dark ring-form trophozoites and chromatin dots. In uninfected erythrocytes, activations remain diffuse across the cytoplasm, demonstrating that predictions are rooted in pathological hallmarks.

B. Inspection of Misclassifications

To critically assess error modes, we analyzed the 348 misclassified test images (4.21% error rate):

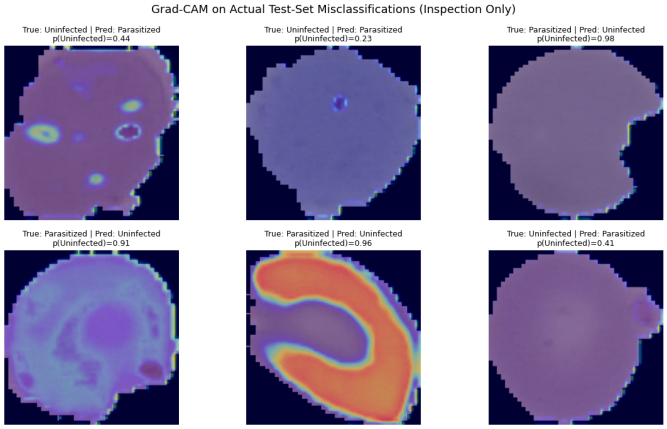


Fig. 6. Grad-CAM heatmaps on test-set misclassifications revealing slide artifacts, central pallor ambiguity, and faint staining.

As illustrated in Fig. 6, errors predominantly stem from:

1) Staining Precipitation & Debris: Platelet superimposition or stain crystallization mimicking parasite chromatin.
2) Atypical Central Pallor: Erythrocyte membrane folds creating ring-like optical contrast in healthy cells.
3) Faint/Early Ring Stages: Low-contrast early ring stages with subtle staining intensity.

C. LIME and SHAP Local Explanations

To complement Grad-CAM, LIME and Kernel SHAP were applied to generate model-agnostic feature attributions.

In Fig. 7, LIME superpixel segment boundaries tightly isolate the physical parasite within the cell membrane. Similarly, SHAP positive SHAPley values align with the dark inclusion bodies, confirming convergent validity across distinct interpretability paradigms.

VI. Conclusion

This study presented a rigorous, leakage-free deep learning system for automated malaria diagnosis from Giemsa-stained thin blood smear cell images. By enforcing strict 70:30 dataset isolation, 3-fold stratified cross-validation, and targeted regularization, the custom CNN achieved **95.79**% test accuracy and an ROC-AUC of **0.9831** on 8,268 untouched test images. Crucially, the model demonstrated a high parasitized recall of **95.31**%, significantly reducing false negative risks. Multi-modal explainability

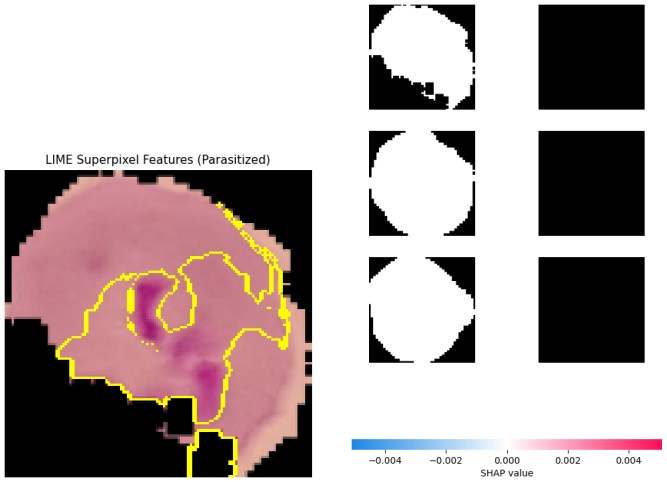


Fig. 7. (Left) LIME superpixel segment boundary identifying parasite inclusion; (Right) SHAP gradient-based feature importance.

through Grad-CAM, SHAP, and LIME corroborated that predictions rely on genuine biological markers of malaria infection. The system exhibits strong potential for deployment in resource-constrained microscopic screening workflows.

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